Lab 13-B

Electric Network Stability & Synthesis

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13.1 Objective

To get familiarize with stabilization of a RLC Network by manipulating poles and zeros and Synthesizing the Electric Networks.

13.2 Introduction

This Lab will walk you through the analysis of a Electric Circuit's response and the impact of poles and zeros on the behavior and response of a linear time-invariant (LTI) system. We'll learn how to stabilize n Electric Circuit by manipulating its poles and zeros. Stability is crucial for a system's performance and reliability in engineering. Through hands-on exploration, we'll master the techniques used to achieve stability in dynamic systems. So, let's get started and uncover the secrets behind stabilizing LTI systems!

13.3 Transfer Function

A transfer function is a mathematical representation of the relationship between the input and output of a linear time-invariant (LTI) system. It describes how the system's output responds to an input signal over a range of frequencies.

Mathematically, a transfer function is typically denoted by H(s) in the Laplace domain. In the Laplace domain, 's' represents the complex frequency variable.

For continuous-time systems in the Laplace domain, the transfer function is defined as the ratio of the Laplace transform of the output (N(s)) to the Laplace transform of the input (D(s)), assuming zero initial conditions:

$$H(s) = \frac{N(s)}{D(s)}$$

13.4 Poles and Zeros

Poles: Poles are the values of 's' (complex numbers) that make the denominator of the transfer function equal to zero i.e., $D(s) = 0$. In other words, they are the roots of the denominator polynomial. Poles determine the dynamic behavior of the system. The location of poles in the complex plane directly affects stability and transient response. Stable systems have poles with negative real parts, while unstable systems have poles with positive real parts. They are called poles since they cause $H(s) \rightarrow \infty$.

Zeros: Zeros are the values of 's' that make the numerator of the transfer function equal to zero i.e., $N(s) = 0$. They are the roots of the numerator polynomial. Zeros represent the frequencies at which the system's output has a null response. They influence the frequency response and play a crucial role in shaping the system's behavior. They are called zeros because they make $H(s) = 0$.

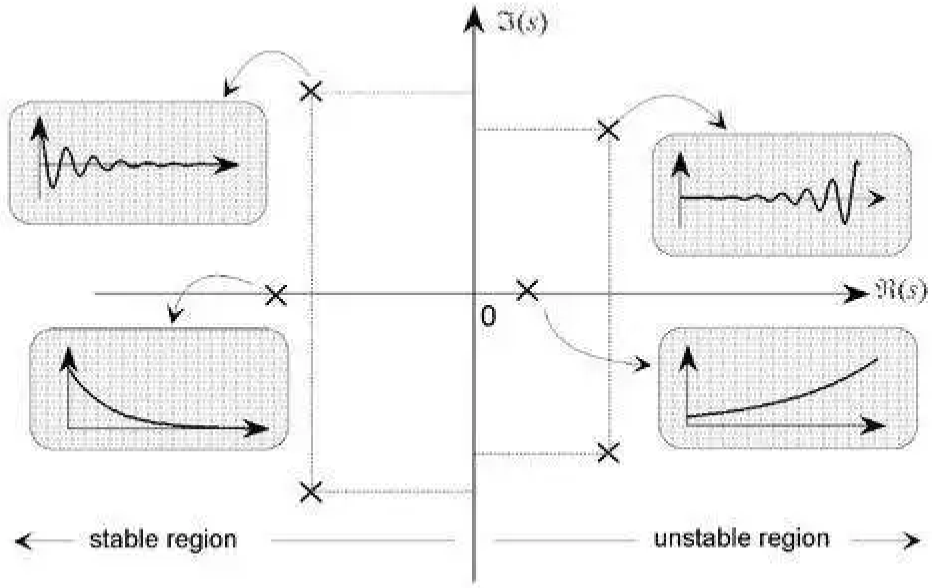


Figure 13.1: Pole and zero locations and the corresponding transient response

13.5 Network Stability

Stability refers to the property of a system where its output remains bounded or converges to a steady-state value over time when subjected to certain inputs or disturbances. A stable control system is desirable because it ensures that the system's response does not grow uncontrollably, leading to erratic or unpredictable behavior.

There are several types of stability in control systems, we are going to discuss.

13.5.1 Bounded Input Bounded Output

"Bounded Input Bounded Output" or BIBO, refers to the behavior of a system in response to different inputs. A system is said to be BIBO stable if, for any bounded input signal, the resulting output signal remains bounded as well. In simpler terms, BIBO stability ensures that the output of a system doesn't grow indefinitely when subjected to a finite or bounded input. Figure 13.1 shows the response of systems and location of its poles represented by "x".

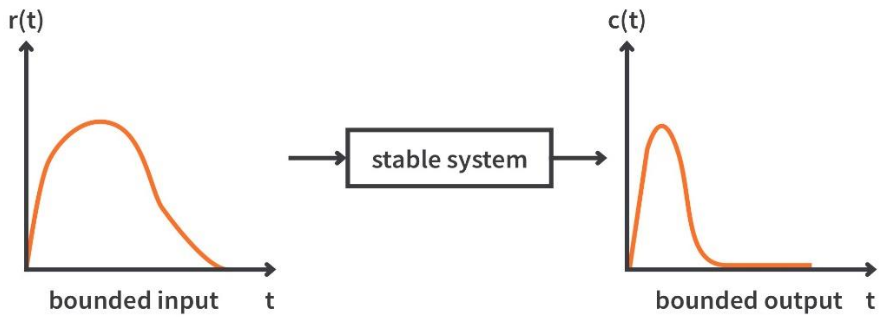


Figure 13.1 Example of a stable system

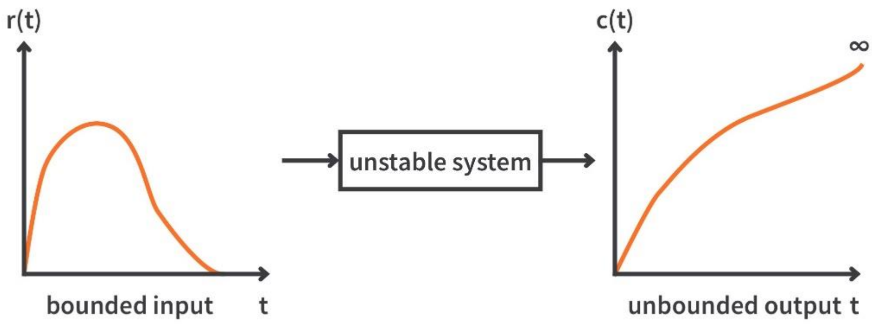


Figure 13.2 Example of an unstable system

The stability of a system with reference to BIBO is summarized as:

1. A system is stable if every bounded input yields a bounded output. Example of such system is shown in Figure 13.1
2. A system is unstable if any bounded input yields an unbounded output. Example of such system is shown in Figure 13.2

13.5.2 The Pole-Zero Plot

A system is characterized by its poles and zeros in the sense that they allow reconstruction of the input/output differential equation. In general, the poles and zeros of a transfer function may be complex, and the system dynamics may be represented graphically by plotting their locations on the complex s-plane, whose axes represent the real and imaginary parts of the complex variable s. Such plots are known as pole-zero plots. It is usual to mark a zero location by a circle (o) and a pole location a cross (×). The location of the poles and zeros provide qualitative insights into the response characteristics of a system.

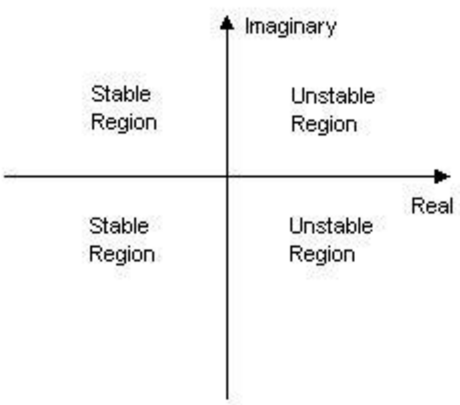


Figure 13.3 System Stability according to the poles of system

The stability of a system can be determined from its poles:

i. Stable systems have poles only in the left-hand plane.
ii. Unstable systems have at least one pole in the right-hand plane and/or poles of multiplicity greater than 1 on the imaginary axis.
iii. Marginally stable or critically stable systems have one pole on the imaginary axis and the other poles in the left-hand plane.

The Figure 13.4 gives examples of poles of stable, unstable and marginally/critically stable systems.

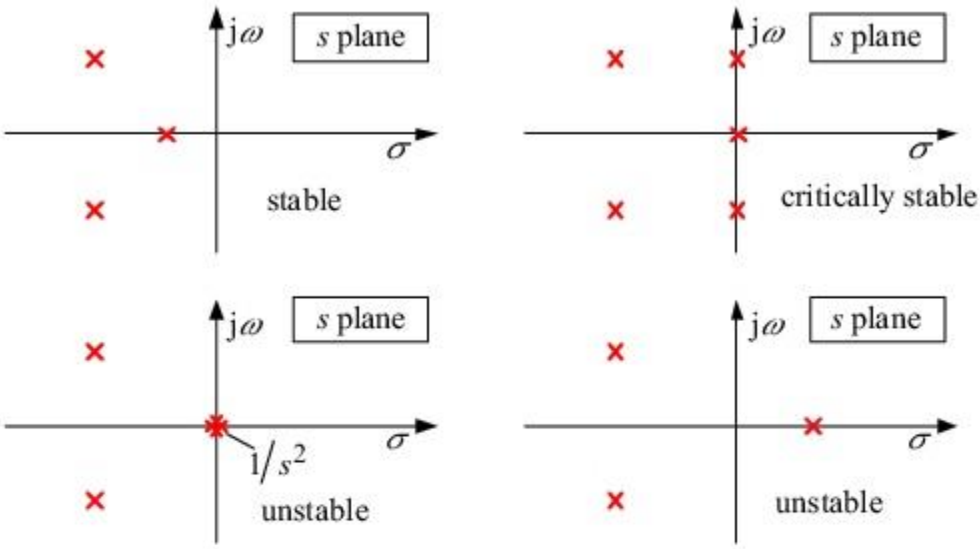


Figure 13.4: Example of poles of different systems

13.5.3 Stability Criteria

Transfer Function H(s) must meet two requirements for the circuit to be stable.

1. The degree of N(s) must be less than the degree of D(s)
2. All the poles of its transfer function H(s) must have negative real parts; in other words, all the poles must lie in the left half of the s-plane.

An unstable circuit never reaches steady state because the transient response does not decay to zero. Consequently, steady-state analysis is only applicable to stable circuits.

13.6 Network Synthesis

Network synthesis may be regarded as the process of obtaining an appropriate network to represent a given transfer function. Network synthesis is easier in the s-domain than in the time domain.

In network analysis, we find the transfer function of a given network. In network synthesis, we reverse the approach: Given a transfer function, we are required to find a suitable network. *Keep in mind that* *in synthesis, there may be many different answers—or possibly no answers—because there are many circuits that can be used to represent the same transfer function; in network analysis, there is only one answer.*

Task 3: Network Synthesis

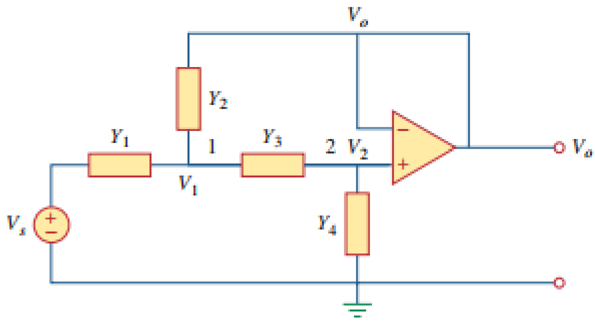


Figure 13.6

a) Synthesize the transfer function provided below by applying Nodal Analysis,

$$T(s) = \frac{V_o}{V_s} = \frac{10^6}{s^2 + 100s + 10^6}$$

Since the transfer function provided is second order, therefore, it must have two capacitors or two inductors or one inductor and one capacitor. Let's keep it simple and consider Y1 and Y3 to be resistors R1 and R2, and Y2 and Y4 are capacitors C1 and C2.

Since, $V = IZ$

$$I = \frac{V}{Z} \Rightarrow \left\{ Y = \frac{1}{Z} \right\}$$

$$I = VY$$

at Node V_1:

$$(V_s - V_1)Y_1 = Y_2(V_1 - V_0) + Y_3(V_1 - V_0)$$

$$Y_1V_s - Y_1V_1 = Y_2V_1 - Y_2V_0 + Y_3V_1 - Y_3V_0$$

$$Y_1V_s = Y_2V_1 - Y_2V_0 + Y_3V_1 - Y_3V_0 + Y_1V_1$$

$$V_s = \frac{Y_2V_1 - Y_2V_0 + Y_3V_1 - Y_3V_0 + Y_1V_1}{Y_1}$$

at Node V_0/V_2:

$$Y_3(V_0 - V_1) + Y_4V_0 = 0$$

$$Y_3V_0 - Y_3V_1 + Y_4V_0 = 0$$

$$V_0(Y_3 + Y_4) = Y_3V_1$$

$$\boxed{V_0 = \frac{Y_3V_1}{Y_3 + Y_4}}$$

$$T(s) = \frac{V_o}{V_s}$$

$$= \frac{y_3 V_1}{y_3 + y_4} \times \frac{y_1}{y_2 V_1 - y_2 V_o + y_3 V_1 - y_3 V_o + y_1 V_1}$$

$$= \frac{y_3 V_1}{y_3 + y_4} \times \frac{y_1}{y_2 V_1 - y_2\left(\frac{y_3 V_1}{y_3 + y_4}\right) + y_3 V_1 - y_3\left(\frac{y_3 V_1}{y_3 + y_4}\right) + y_1 V_1}$$

$$= \frac{y_3 V_1}{\cancel{y_3 + y_4}} \times \frac{y_1 \cancel{(y_3 + y_4)}}{\cancel{y_2 y_3 V_1} + y_2 y_4 V_1 - \cancel{y_2 y_3 V_1} + \cancel{y_3^2 V_1} + y_3 y_4 V_1 - \cancel{y_3^2 V_1} + y_1 y_3 V_1 + y_1 y_4 V_1}$$

$$T(s) = \frac{y_1 y_3}{y_2 y_4 + y_3 y_4 + y_1 y_3 + y_1 y_4}$$

Since; $y_1 = \frac{1}{R_1}$, $y_3 = \frac{1}{R_2}$, $y_2 = sC_1$, $y_4 = sC_2$

$$T(s) = \frac{\frac{1}{R_1} \cdot \frac{1}{R_2}}{s^2 C_1 C_2 + \frac{sC_2}{R_2} + \frac{1}{R_1 R_2} + \frac{sC_2}{R_1}}$$

$$= \frac{1/\cancel{R_1 R_2}}{(s^2 R_1 R_2 C_1 C_2 + sR_1 C_2 + 1 + sR_2 C_2)/\cancel{R_1 R_2}}$$

$$T(s) = \frac{1/R_1 R_2 C_1 C_2}{s^2 + s\left(\frac{R_1 + R_2}{R_1 R_2 C_1}\right) + \frac{1}{R_1 R_2 C_1 C_2}}$$

Page No.

b) Once you are done with synthesis Vo/Vs (in terms of R1 R2 C1 and C2), compare the equation with the Transfer function provided. And find out the Values of R1, R2, C1 and C2. Assumptions will be done.

Comparing poles:

$$\frac{R_1 + R_2}{R_1 R_2 C_1} = 100 \quad \text{— (i)}$$

Comparing zeroes:

$$\frac{1}{R_1 R_2 C_1 C_2} = 10^6 \quad \text{— (ii)}$$

Assume $R_1 = 10k\Omega$, $R_2 = 20k\Omega$, and use them in eq (i).

$$\cancel{\frac{10k + 20k}{200k\,C_1} = 100}$$

$$\cancel{30k}$$

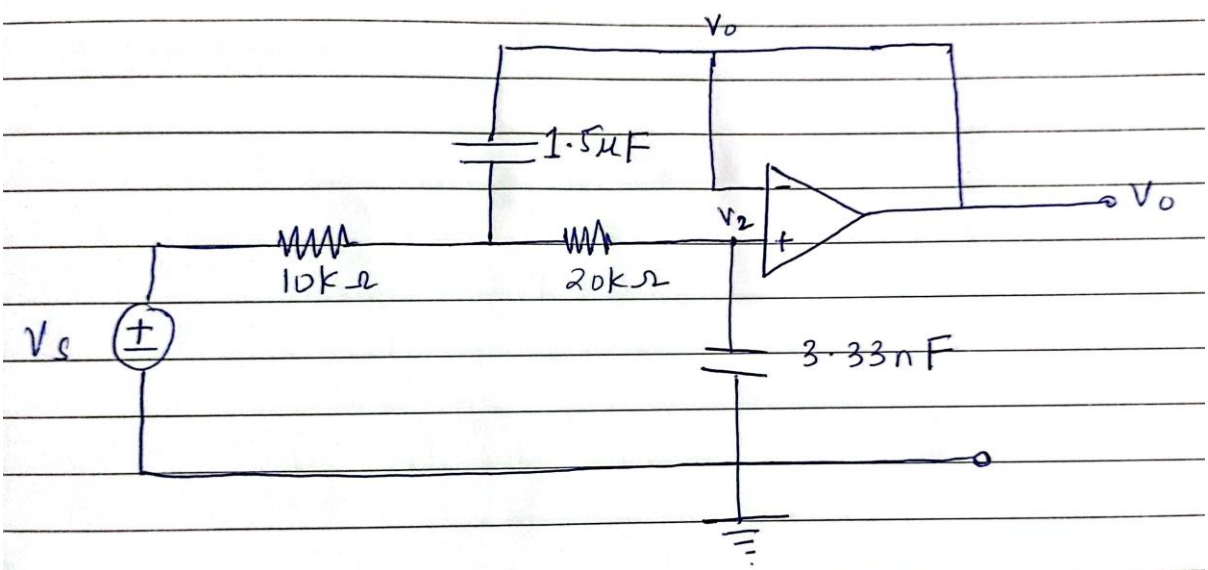
$$\frac{30k}{(20k)(10k)(100)} = C_1$$

$$\frac{30 \times 10^3}{200000 \times 10^6} = \boxed{C_1 = 1.5\mu F}$$

$$\frac{1}{(10k)(20k)(1.5\mu)\,C_2} = 10^6$$

$$\frac{1}{(10k)(20k)(1.5\mu)(10^6)} = \boxed{C_2 = 3.33nF}$$

c) Redraw the synthesized circuit and annotate the schematic with values of resistors and the capacitors.



Assessment Rubric
Lab 13-B

Name:	Student ID:

Points Distribution

Task No.	LR 2 Simulation	LR6 Calculations	LR5 Results/Plots	LR 10 Analysis	LR 11 Design
Task 1	-	/10	/10	/5	
Task 2	/5	/10	/15	/10	/5
Task 3	-	/7.5	/5		/7.5
Total	/5	/35	/30	/20	
CLO Mapped	CLO 2	CLO 2	CLO 2	CLO 2	

Affective Domain Rubric		Points	CLO Mapped
AR 7	Report Submission	/10	CLO 4

CLO	Total Points	Points Obtained
2	90	
4	10	
Total	**100**	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3:Good Points 3	Level 4:Exemplary Points 4
LR1	**Circuit Layout**	Connections between circuit components are mostly wrong. Circuit layout is cluttered. Needs guidance to make correct equipment/ component connections.	Few of the connections made between circuit components are wrong. Circuit layout is not neat and clean as per standards.	Circuit layout and connections are correct but not all connections are neat and clean as per standards.	Neat, clean and correct connections of all the circuit components/ equipment are made as per standard circuit diagram.
LR2	**Program/Code/ Simulation Model/ Network Model**	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR3	**Troubleshooting**	Unable to identify the fault/minimal effort show in troubleshooting.	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and takes necessary steps and actions to correct it.
LR4	**Data Collection**	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included.	Measurements are somewhat inaccurate and imprecise. Observations are incomplete or vague. Major errors are there in using symbols, units and significant digits.	Measurements are mostly accurate. Observations are generally complete. Minor errors are present in using symbols, units and significant digits.	Measurements are both accurate and precise. Data collection is systematic. Observations are very thorough and include appropriate symbols, units and significant digits and task completed in due time.
LR5	**Results & Plots**	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR6	**Calculations**	Formulae used and/or calculations are incorrect. Units are not mentioned with results.	Formulae used are correct but there are few mistakes in calculation which lead to incorrect results.	Formulae used and end results are correct. Complete working steps are not shown. Proper units are not mentioned with results.	Formulae used, end results and all calculations are correct with all the intermediate steps clearly shown. Units are mentioned with results.
LR9	**Report**	All the in-lab tasks are not included in report and / or the report is submitted too late.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included. Report is submitted after due date.	Good summary of most the in-lab tasks is included in report. The work is supported by figures and plots with explanations. The report is submitted timely.	Detailed summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots and tables.
LR10	**Analysis**	Results are not interpreted or are analyzed incorrectly.	Analysis of the obtained results is incomplete. Conclusions are not drawn.	Results are analyzed and concluded but are not well explained and justification is not provided with the help of reasoning.	Results are well analyzed supported with reasons. Good comparison is made between the theoretical and experimental results with correct and useful conclusion.
LR11	**Design**	Proposed design is unsubstantiated or does not satisfy most of given constraints. The breakdown of system into features is incomplete and still at lower resolution.	Justification for proposed design is weak, and it only satisfies some constraints. The breakdown of system is missing some features and resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies most given constraints. The breakdown of system into features seems complete, but resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies all given constraints. The breakdown of system into features seems complete and at appropriate resolution, given students' level.